AI UNIT – 1

Introduction to Artificial Intelligence (AI)

1. Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a **branch of computer science** that **focuses** on **creating systems** capable of **performing** tasks that normally **require human intelligence**.

These tasks include **learning**, **reasoning**, **problem-solving**, **perception**, **understanding** natural language, and **decision-making**.

Definition:
Artificial Intelligence can be defined as *the **study** and **development** of **computer systems** that can **perceive** their **environment**, **think**, **learn** from experience, and **take actions** that maximize the chances of achieving goals.*

AI aims to **simulate human intelligence** in machines so that they can **function autonomously** and **intelligently**.

Key characteristics of AI systems:

- **Ability to learn** from data and experience
- **Reasoning** and **problem-solving** capability
- **Adaptability** to new situations
- **Decision-making** under uncertainty
- **Interaction** with humans using natural language

What is Artificial Intelligence?

According to the father of Artificial Intelligence, **John McCarthy**, it is "The **science** and **engineering** of **making intelligent machines**, especially intelligent computer programs".

Artificial Intelligence is a **way of making a computer**, a computer-controlled robot, or a software **think intelligently**, in the similar manner the intelligent humans think.

AI is accomplished by studying how human brain thinks, and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

Philosophy of AI

While exploiting the power of the computer systems, the curiosity of human, lead him to wonder, "**Can a machine think and behave like humans do?**"

Thus, the development of AI started with the intention of creating similar intelligence in machines that we find and regard high in humans.

Goals of AI

- To **Create Expert Systems** − The systems which exhibit **intelligent behaviour**, **learn**, **demonstrate**, **explain**, and **advice** its users.
- To Implement Human Intelligence in Machines − Creating systems that **understand**, **think**, **learn**, and **behave** like **humans**.
- **Replicate** human intelligence
- Solve **Knowledge-intensive** tasks
- An **intelligent connection** of **perception** and **action**
- Building a machine which can perform tasks that requires human intelligence such as:
  - **Proving** a theorem
  - **Playing** chess
  - **Plan** some surgical operation
  - **Driving** a car in traffic
- Creating some system which can exhibit intelligent behaviour, **learn new things by itself**, demonstrate, explain, and can advise to its user.

Why Artificial Intelligence?

Before Learning about Artificial Intelligence, we should know that what is the **importance of AI and why should we learn it**? Following are some main reasons to learn about AI:

- With the help of AI, you can create such software or devices which can **solve real-world problems** very easily and with accuracy such as health issues, marketing, traffic issues, etc.

- With the help of AI, you can create your **personal virtual Assistant**, such as Cortana, Google Assistant, Siri, etc.
- With the help of AI, you can build such **Robots** which **can work** in an **environment** where **survival of humans** can be at **risk**.
- AI opens a path for other **new technologies**, new devices, and new Opportunities.

What Contributes to AI?

Artificial intelligence is a science and technology **based on disciplines** such as **Computer Science**, **Biology**, **Psychology**, **Linguistics**, **Mathematics**, and **Engineering**.

A major thrust of AI is in the development of computer functions associated with human intelligence, such as reasoning, learning, and problem solving.

Out of the following areas, one or multiple areas can contribute to build an intelligent system.

What is AI Technique?

In the real world, the **knowledge** has some **unwelcomed properties** −

- **Volume** is **huge**, next to unimaginable.
- It's is **not well-organized** or well-formatted.
- It keeps **changing constantly**.

AI Technique is a manner to **organize and use the knowledge efficiently** in such a way that −

- It should be **perceivable** by the people who provide it.
- It should be easily **modifiable** to correct errors.
- It should be **useful in many situations** though it is **incomplete or inaccurate.**

AI techniques elevate the speed of execution of the complex program it is equipped with.

History of Artificial Intelligence (20th Century)

Artificial Intelligence did not appear suddenly — it **evolved slowly** through **ideas**, **experiments**, and **inventions across the 20th century**.

The journey of AI is a fascinating story of imagination turning into science and science turning into technology.

1923 – Birth of the Word "Robot"

Karel Čapek's play "Rossum's Universal Robots (RUR)" opened in London. This play introduced the word **"robot"** into the **English language for the first time.**
It sparked global imagination about artificial beings and intelligent machines.

1943 – Foundations of Neural Networks

The **theoretical foundations of neural networks** were laid.
This work introduced the **idea that machines could mimic** the structure of the **human brain** using artificial neurons — a concept that later became central to modern AI and deep learning.

1945 – Birth of Robotics

Isaac Asimov, a Columbia University alumnus, **coined the term "Robotics"**.
He also proposed ethical laws for robots (Three Laws of Robotics), shaping early thinking about machine intelligence and responsibility.

1950 – Intelligence as a Scientific Question

Two major milestones occurred:

- Alan Turing introduced the Turing Test and published *"Computing Machinery and Intelligence"*, asking the famous question:
  "Can machines think?"
- Claude Shannon published work on chess-playing programs, treating intelligence as a search and decision-making process.

This year transformed intelligence into a scientific research problem.

1956 – Birth of Artificial Intelligence as a Field

John McCarthy officially coined the term **"Artificial Intelligence"**.
The first AI program was demonstrated at **Carnegie Mellon University**.

This year is widely considered the **formal birth of AI as a discipline**.

1958 – First AI Programming Language

John McCarthy developed **LISP**, the **first programming language** specially designed **for AI** research.
LISP became the backbone of AI development for decades.

1964 – Natural Language Understanding

Danny Bobrow (MIT) showed that **computers** could **understand natural language** well enough to solve algebra word problems.
This proved that machines could process human language meaningfully.

1965 – First Chatbot (ELIZA)

Joseph Weizenbaum developed **ELIZA**, the **first conversational program**.
It could interact in English and simulate human conversation — an early ancestor of today's chatbots like ChatGPT.

1969 – First Intelligent Robot (Shakey)

Scientists at Stanford Research Institute built Shakey, the first robot with:

- movement
- perception
- reasoning
- problem-solving abilities

Shakey was the world’s first intelligent autonomous robot.

1973 – Vision-Based Robotics

Edinburgh University built Freddy the Robot, capable of:

- seeing objects
- identifying them
- assembling models

This was a major step in computer vision and robotics.

1979 – Autonomous Vehicles Begin

The **Stanford Cart**, the first **computer-controlled autonomous vehicle**, was developed.
This marked the beginning of self-driving vehicle research.

1985 – Creative AI

Harold Cohen created AARON, a drawing program that produced original artwork.
This showed that AI could be creative, not just logical.

1990s – Expansion Era of AI

AI research expanded rapidly into multiple fields:

- Machine learning
- Case-based reasoning
- Multi-agent systems
- Scheduling systems
- Data mining
- Web crawlers

- Natural language translation
- Computer vision
- Virtual reality
- AI in games

AI moved from laboratories to **real-world applications**.

1997 – AI Beats Human World Champion

IBM's Deep Blue defeated **world chess champion Garry Kasparov**. This became one of the most iconic moments in AI history, proving that machines could outperform humans in complex strategic tasks.

2000 – AI Enters Daily Life

Major developments made AI visible to the public:

- **Robot pets** became commercially available
- **MIT's Kismet** robot showed emotional expressions
- **Nomad robot** explored Antarctica and located meteorites

AI moved from research labs into **homes, society, and real environments**.

Advantages of Artificial Intelligence

- **High Accuracy with less errors**: AI machines or systems are prone to less errors and high accuracy as it takes decisions as per pre-experience or information.
- **High-Speed**: AI systems can be of very high-speed and fast-decision making, because of that AI systems can beat a chess champion in the Chess game.
- **High reliability**: AI machines are highly reliable and can perform the same action multiple times with high accuracy.
- **Useful for risky areas**: AI machines can be helpful in situations such as defusing a bomb or exploring the ocean floor, where employing a human can be risky.

- **Digital Assistant**: AI can be very useful to provide digital assistant to the users such as AI technology is currently used by various E-commerce websites to show the products as per customer requirement.
- **Useful as a public utility**: AI can be very useful for public utilities such as a **self-driving car** which can make our journey safer and hassle-free, **facial recognition** for security purpose, **Natural language processing** to communicate with the human in human-language, etc.

Disadvantages of Artificial Intelligence

- **High Cost**: The hardware and software requirement of AI is very costly as it requires lots of maintenance to meet current world requirements.
- **Can't think out of the box**: Even we are making smarter machines with AI, but still they cannot work out of the box, as the robot will only do that work for which they are trained, or programmed.
- **No feelings and emotions**: AI machines can be an outstanding performer, but still it does not have the feeling so it cannot make any kind of emotional attachment with human, and may sometime be harmful for users if the proper care is not taken.
- **Increase dependency on machines**: With the increment of technology, people are getting more dependent on devices and hence they are losing their mental capabilities.
- **No Original Creativity**: As humans are so creative and can imagine some new ideas but still **AI machines cannot beat this power of human intelligence** and **cannot be creative and imaginative**.

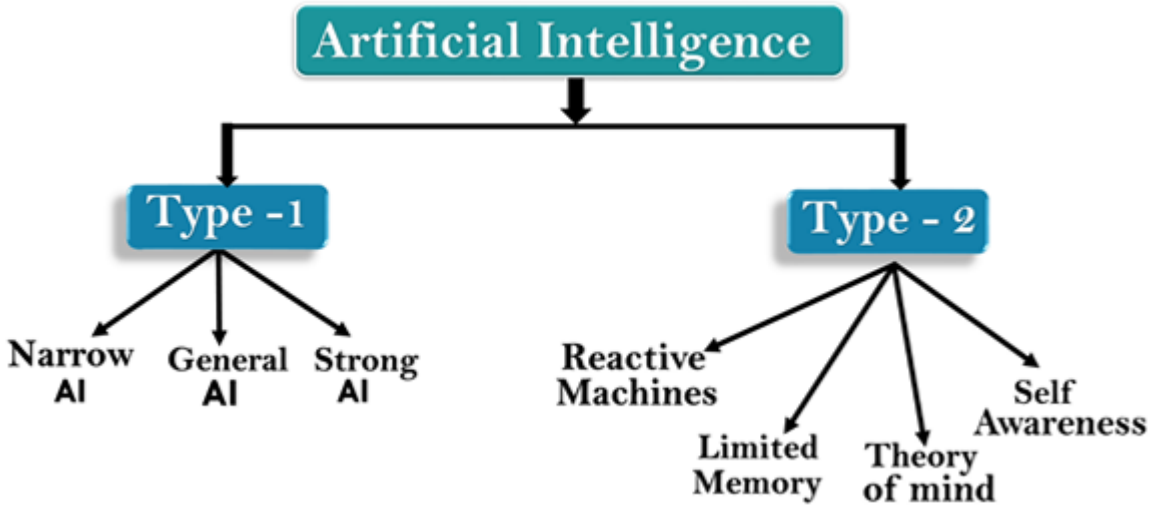
Intelligent Systems: Categorization of Intelligent System

Intelligent Systems

An **Intelligent System** is **a system that can perceive its environment**, **interpret it**, **learn from experience**, and **act autonomously**.

Artificial Intelligence can be divided in various types, **there are mainly two types of main categorization** which are **based on capabilities** and based on **functionality** of AI.

Following is flow diagram which explain the types of AI.



AI type-1: Based on Capabilities

1. Weak AI or Narrow AI:

- Narrow AI (Weak AI) is **designed to perform specific tasks** using intelligence.
- Most AI systems **available today** are **Narrow AI**, as they are **built for limited applications.**
- It **cannot operate outside its trained domain**; hence it is also called **Weak AI**.
- If it is **used beyond its limitations**, it may **fail unpredictably**.
- **Examples**: Apple Siri (limited voice assistant),

  IBM Watson (domain-specific expert system),

  chess-playing AI,

  recommendation systems,

  self-driving cars,

  speech recognition,

  image recognition.

2. General AI:

General AI (Strong AI) refers to a **system that can perform** any **intellectual task with human-like efficiency.**

- The goal of General AI is to develop **machines that can think and reason independently**, similar to humans.
- Currently, **no system exists** that can **perform** all tasks **as effectively as a human.**
- **Researchers** worldwide are **actively working** on developing General AI.
- Since General AI is still in research, it **will require significant time** and **effort** to achieve.

3. Super AI:

- **Super AI** refers to a level of artificial intelligence where **machines surpass human intelligence** and **outperform humans in all cognitive** tasks.
- It is considered an **advanced stage beyond General (Strong) AI**, with **abilities** such as **reasoning**, **learning**, **planning**, **judgment**, and autonomous **decision**-making.
- **Super AI remains a hypothetical concept**, and **developing** such systems in reality would be a **world-changing** and **highly challenging** task.

Artificial Intelligence type-2: Based on functionality

1. Reactive Machines

- Purely reactive machines are the most **basic type of artificial intelligence**.
- These systems **do not store memories or use past experiences** to influence **future actions**.
- They **operate** solely based **on the current input** and **select actions** accordingly.
- **Examples** -

IBM's Deep Blue is a classic example of a reactive machine.

- **IBM's Deep Blue** was a **chess-playing computer system** developed by **IBM**.
- It became famous in **1997** when it **defeated the world chess champion Garry Kasparov** in a six-game match — a major milestone in AI history.
- It could evaluate **about 200 million chess positions per second**

- Used **brute-force search**, advanced algorithms, and **hand-crafted evaluation functions**

- Relied heavily on **human chess expertise** programmed into the system

- Play chess at a **superhuman level**
- **Analyse huge numbers of possible moves** very quickly
- **Choose** the statistically **best move** based on calculations

What Deep Blue could NOT do:

- **Learn from experience** like modern machine learning systems
- **Play** any **game other than chess**
- **Think**, **reason**, or **understand chess** like a human
- **Transfer** its intelligence to other tasks

ELIZA is an example of a reactive AI system that operates using predefined rules without learning from past interactions.

- ELIZA was an early computer program created in **1966**.
- It's often described as the **first chatbot** (or at least the first famous one). What made it special wasn't intelligence in the modern sense, but how *convincing* it felt to people.

- What ELIZA did -

- ELIZA worked by using **pattern matching and simple rules** to transform what a user typed into a response.

  So, if you said:

  "I feel unhappy."

  ELIZA might reply:

  "Do you often feel unhappy?"

- It didn't *understand* emotions—it just:
  - Looked for **keywords**
  - **Rephrased** your input
  - Asked **open-ended questions**

  That was enough to keep a conversation going.

2. Limited Memory

- Limited memory machines can **store past experiences or data** for a **short period** of time.
- These systems use **stored information temporarily** to make decisions.

- **Self-driving cars** are a common **example** of limited memory AI, as they use recent data such as the speed and distance of nearby vehicles and traffic signals to navigate safely.

3. Theory of Mind

- **Theory of Mind AI** refers to a type of artificial intelligence that is designed to **understand human mental states**, such as **emotions**, **beliefs**, **intentions**, and **thoughts**.
- Just like humans **consider what others are feeling or thinking before responding**, this type of AI would be able to do the same.
- This type of AI systems-
    - Recognize and interpret **human emotions** (happiness, sadness, anger, fear, etc.)
    - Understand that **different people have different beliefs, knowledge, and intentions**
    - **Predict** how humans might behave based on their emotional and mental states
    - **Interact socially in a natural and empathetic way**, similar to human-to-human communication
- At present, **Theory of Mind AI does not exist**.
- Today's AI systems can recognize patterns, analyse data, and even simulate conversation, but they **do not truly understand emotions or beliefs**.
- They lack genuine awareness of mental states.

4. Self-Awareness

- **Self-Awareness AI does not exist**.
- It remains a **theoretical and hypothetical concept**.
- In this type of system **machines would possess consciousness, self-awareness, and emotions**, similar to or even beyond those of humans.
- Such AI systems would **not only perform tasks or understand others** but would also be **aware of their own existence, internal states, and thoughts**.
- These machines would be:

    - **Super intelligent**, exceeding human intelligence in reasoning, learning, and decision-making
    - Capable of having **sentiments, emotions, and feelings**
    - **Aware of themselves** as independent entities, not just tools or programs

- A self-aware AI could understand concepts like *"I exist," "I am thinking,"* and *"I feel."* This goes far beyond current AI capabilities.

Components of Artificial Intelligence (AI)

- **Key Components of AI**
  The **main parts of AI that work together** to make the system intelligent are as follows:

1. Learning in AI

- Learning in Artificial Intelligence (AI) refers to the **process by which an AI system improves its performance** over time **by analysing data**, **identifying patterns** and making **decisions** based on **past experience**.
- It helps them to adapt and change without being explicitly programmed for every task.
- **Types of Learning:**

1. Supervised Learning:

- Supervised learning is a type of machine learning where the **model is trained using labelled data** (input data with correct output).
- The algorithm learns by **comparing its predictions with the actual answers** and **correcting errors**.
- It is mainly used to **make predictions based on known input-output relationships**.
- Common examples include **spam email detection, image recognition, and price prediction**.

2. **Unsupervised Learning**:

- Unsupervised learning is a type of machine learning where the **model is trained using unlabelled data** (no correct answers are provided).
- The algorithm tries to **find hidden patterns, structures, or relationships** in the data on its own.
- It **groups data based on similarities or relationships** without prior knowledge of categories.
- Common examples include **customer segmentation, market basket analysis, and grouping similar images**.

3. **Reinforcement Learning**:

- Reinforcement learning (RL) is a type of machine learning where an **agent learns by interacting with an environment**.
- The **agent receives rewards or penalties based on its actions** and learns to maximize cumulative rewards.
- It is mainly used for **decision-making and sequential tasks** where actions affect future outcomes.
- Common examples include **self-driving cars, game-playing AI (like Chess or Go), and robotic navigation**.

4. **Semi-Supervised Learning**:

- Semi-supervised learning is a type of machine learning where the **model is trained** using **a small amount of labelled data and a large amount of unlabelled data**.
- It combines the advantages of **supervised and unsupervised learning**.
- The algorithm **learns patterns from unlabelled data** while being guided by the labelled data.
- Common examples include **web content classification, medical image analysis, and speech recognition**.

5. **Self-Supervised Learning**:

- Self-supervised learning is a type of machine learning where the **model generates its own labels from the input data**.
- It does **not require manually labelled data**, saving time and effort in data preparation.
- The algorithm **learns by predicting part of the data from other parts**, creating tasks from the data itself.
- Common examples include **natural language processing (like GPT), image representation learning, and speech processing**.

6. **Deep Learning**:

- Deep learning is a subset of machine learning that uses **artificial neural networks with multiple layers** to learn from data.
- It can automatically **extract features and patterns** from raw data without manual feature engineering.

- Deep learning is especially **powerful for complex tasks** like image recognition, speech recognition, and natural language processing.
- Common examples include **self-driving cars, virtual assistants (like Siri), and facial recognition systems**.

2. Reasoning in AI

Reasoning **helps AI to make logical decisions**, **solve problems** and **infer conclusions** based on given information much like human reasoning.

It allows AI to use information and rules to come to conclusions just like we humans do.

Types of Reasoning:

1. Deductive Reasoning: **Drawing conclusions based on general rules** or **premises**. Used in rule-based systems and expert systems for logical decision-making**.**
Example:
- **Premise:** All exams start at 9 AM
- **Premise:** There is an exam today.
- **Conclusion:** Today's exam starts at 9 AM

2. Inductive Reasoning: Making **general conclusions from specific examples**. Found in machine learning where models generalize from training data**.**
Example:
- **Observation:** Every time I studied regularly, I passed the exam.
- **Conclusion:** Studying regularly will help me pass future exams.

3. Abductive Reasoning: **Finding the best explanation** for a given set of observations. Used in diagnostic systems like medical diagnosis tools.
Example:
- Observation: The lawn is wet.
- Possible explanation: It rained last night.

AI reasoning is important for applications like financial analysis where system predicts market trends based on historical data.

3. Ethics and Bias in AI

Systems learn from data but the data they use can have biases. It's important to mention that AI systems need to be carefully designed and monitored to avoid unfair biases or making unethical decisions.

- **Ethical Considerations**: Systems **must be designed responsibly to ensure they do not harm individuals or groups** and their decisions should be transparent.
- **Bias in AI**: If the **training data is biased AI might make biased predictions** (e.g. racial bias in facial recognition software).

4. Problem Solving in AI

Problem-solving in AI **helps it in identifying complex problems and finding ways to solve** them efficiently. System use algorithms designed to solve problems across various domains:

1. **Search Algorithms:** Finding the best way to solve a problem such as A* for path finding in maps or games.
3. **Optimization Techniques:** Find the **best solution from a set of possible solutions.**
4. **Simulation and Modelling:** Used in cases where **real-world testing is impractical or expensive** such as climate models or stock market predictions.

It is essential in planning, logistics, automation and the places where complex challenges need to be efficiently and effectively solved.

5. Perception in AI

Perception **allows systems to understand their surroundings** by processing sensory inputs such as images, sounds and even touch. This is important for AI to interact with the real world.

Main Aspects of AI Perception:

- **Computer Vision:** Helps **AI to interpret and understand visual input** from the surrounding environment. Applications include facial recognition, autonomous vehicles and surveillance systems.
- **Speech Recognition:** Allows AI to **convert spoken language into text**. This is used in virtual assistant technologies, automated customer support and real-time communication devices.

6. Data in AI

Data is considered as the **fuel** that helps AI systems. Accuracy, effectiveness and fairness of AI models are influenced by the quality of the data used to train them. Without good data even the most complex AI algorithms can produce poor or biased results.
Types of Data Used in AI:

1. **Structured Data**: Data which is **organized and easy to analyse** such as data in tables or spreadsheets (e.g. customer information or sales numbers).
2. **Unstructured Data**: This includes data that isn't organized in a pre-defined way such as text, images, videos or social media posts.

7. Language Processing

Language processing which is the part of natural language processing (NLP) **helps machines to read, understand and produce human language**. It involves tasks from basic text analysis to complex conversations.

Core NLP **Functions**:

- **Syntax Analysis:** Understanding the **structure** of sentences.
- **Semantic Analysis: Understands meanings** of sentences which is important for getting context of language.
- **Dialogue Management:** Helps systems to engage in conversation with humans ensuring AI responds correctly.

Foundations of Artificial Intelligence

The foundations of Artificial Intelligence (AI) represent the **core disciplines, theories, and principles** that together enable machines to simulate human intelligence.

AI is not a single subject but a **combination of multiple fields** working together.

Each contributes tools, theories, and methods to make machines “intelligent.”

Mathematics

- **Linear Algebra:** Used in neural networks, representing data as vectors and matrices.
- **Probability & Statistics:** Handles uncertainty, predictions, and Bayesian reasoning.
- **Optimization:** Improves AI models by minimizing error functions or maximizing rewards.

Computer Science

- **Algorithms & Data Structures:** Efficient data processing and searching in AI systems.
- **Programming:** Building AI models using languages like Python, R, or Java.
- **Computational Complexity:** Understanding limits and efficiency of AI solutions.

Cognitive Science / Psychology

- **Human Learning Models:** AI mimics human learning patterns (reinforcement, memory).
- **Decision Making:** Studying human problem-solving to design intelligent agents.
- **Perception & Attention:** Helps AI in vision and natural language understanding.

Linguistics

- **Syntax & Grammar:** Helps AI understand sentence structure (NLP).
- **Semantics:** Understanding meaning of words and sentences.
- **Pragmatics & Context:** Enables AI to interpret language based on situation or context.

Philosophy

- **Logic & Reasoning:** Provides rules for deductive, inductive, and abductive reasoning.
- **Ethics:** Addresses fairness, bias, and responsible AI design.
- **Knowledge Representation:** How AI systems store and process “knowledge.”

Neuroscience / Biology

- **Neural Networks Inspiration:** Modelled after human brain neurons.
- **Learning & Memory Mechanisms:** Helps in reinforcement and deep learning models.
- **Perception:** Vision and auditory models inspired by human senses.

Sub-Areas of Artificial Intelligence

Artificial Intelligence is a broad field composed of several **specialized sub-areas**, each **focusing on a specific type of intelligence**, **capability**, **or application**.

These sub-areas work together to build intelligent systems.

Machine Learning (ML)

- AI systems **learn from data** without being explicitly programmed.
- Includes **Supervised, Unsupervised, and Reinforcement Learning**.
- Used in **spam filters, recommendation systems, predictive analytics**.

Natural Language Processing (NLP)

- Enables computers to **understand, interpret, and generate human language**.
- Includes **syntax analysis, semantic analysis, sentiment analysis, and translation**.
- Used in **chatbots, virtual assistants, and language translation apps**.

Computer Vision

- Enables AI to **interpret images and videos**.
- Tasks include **object detection, face recognition, and image classification**.
- Used in **self-driving cars, surveillance, medical imaging**.

Robotics

- AI applied to **design and control robots** that can sense, act, and learn.
- Includes **motion planning, perception, and manipulation**.
- Used in **manufacturing, delivery robots, and medical surgery robots**.

Expert Systems / Knowledge Representation

- AI systems that **mimic human decision-making** using rules and knowledge bases.
- Includes **reasoning, logic, and inference engines**.
- Used in **medical diagnosis, troubleshooting, and financial advising**.

7. Applications of Artificial Intelligence

1. **Healthcare** – Disease diagnosis, medical imaging, drug discovery
2. **Education** – Intelligent tutoring systems, personalized learning
3. **Finance** – Fraud detection, algorithmic trading
4. **Transportation** – Autonomous vehicles, traffic management
5. **Manufacturing** – Predictive maintenance, robotics
6. **E-commerce** – Recommendation systems, customer support chatbots
7. **Security** – Surveillance, biometric systems

8. Current Trends in Artificial Intelligence

The era witnessed a dramatic increase in tools, applications, and platforms based on AI and Machine Learning (ML). These technologies have impacted healthcare, manufacturing, law, finance, retail, real estate, accountancy, digital marketing, and several other areas.

Companies are investing in AI research to find out how they can bring AI closer to humans. By 2025 AI software revenues alone will reach above $100 billion globally (Figure 1). This means that we will continue seeing the advancement of AI and Machine Learning (ML)-related technology in foreseeable future. AI changes notably fast, so you'll need to go out of your way to keep up with the latest trends if you want to stay as informed as possible. Let's take a look at everything you need to know about the latest AI trends.

1. Intelligent Process Automation

In the latest technology trend, organisations are looking for intelligent automation tools to solve business challenges and increase productivity, efficiency, and accuracy, benefiting the organisation. One of the successive waves, Intelligent Process Automation, or IPA, brings together Robotic Process Automation (RPA) and Artificial Intelligence (AI) technologies to empower rapid end-to-end business process automation and accelerate digital transformation. In RPA, computer software 'robots' handle repetitive, rule-based digital tasks which are driven by structured data. However, many business processes now are fed by or generate large amounts of unstructured and real-time data. IPA makes it possible to automate processes with machine learning and analytic capabilities and cognitive technologies, like computer vision, Natural Language Processing (NLP), and fuzzy logic. The adoption of IPA is expected to grow in the coming days, with large-scale growth expected across several industries.

2. A Shift Toward Cybersecurity

With data becoming more precious than ever before, there's no shortage of cyber criminals out there looking for new ways to compromise it. One of the downsides of novice-level AI is that hackers can manipulate them to access the sensitive information. So, a significant trend in AI is developing technology to recognise and report common types of attacks. Anti-virus software is also being developed by using AI in the same manner as this technology can help prevent a malware threat from having devastating consequences. When it comes to businesses, AI-powered cybersecurity tools also can gather data from a company's own communications networks, digital activity, transactional systems, and websites, as well as other external public sources. These tools then run algorithms to identify patterns and detect or predict threatening activity, potential data breaches, etc. This is a trend we can expect to continuously see in the future as criminals constantly create new malware and data acquisition methods.

3. AI for Personalised Services

As AI becomes more powerful and efficient at researching a particular market and demographic, acquiring consumer data is becoming more accessible than ever. The biggest AI trend in marketing is the increasing focus on providing personalised services. One of the most common ways that AI can do so is through analysing the online activity of individuals who search for specific keywords. This level of personalisation is virtually guaranteed to

provide a better experience for customers, which will directly increase the revenue of companies that take advantage of it. As machine learning becomes more adept at understanding what people want in specific instances, AI will become less of a sales tool and more of a digital friend.

4. Automated AI Development

In coming years, expect to see significant innovations in the area of 'AI for AI': using AI to help automate the steps and processes involved in the life cycle of creating, deploying, managing, and operating AI models. At a certain level, AI can develop its algorithms to solve problems, increase efficiency, and provide humans with useful research data.

Using automated AI will allow even non-experts to use AI algorithms and techniques. One example is Google's AutoML, a tool that simplifies creating machine learning models and makes the technology accessible to a wider audience. These tools can create as much customisation as required without knowing the complex workflow of Machine Learning in detail. Although this type of development is in infancy, automated AI is renowned for growing exponentially and is a major AI trend.

5. Autonomous Vehicles

With companies like Samsung, Nvidia, Volkswagen, Uber, and Google's Waymo, the scope of autonomous driving has increased many folds. Everyone knows AI's functionality into autonomous vehicles, and to tap such immense potential, car, and tech companies are infusing billions of dollars in this domain. The process is driven by the economic and social benefits involved. Car manufacturers hope that autonomous driving technology will sway consumers' minds. Supporters believe that self-driving car technology will reduce traffic deaths and be a safe alternative to drive.

6. Incorporating Facial Recognition

Facial recognition appears to be en vogue at the moment. It is popping up in many aspects of our lives and is being adopted by private and public organisations for various purposes, including surveillance. More countries are readying themselves to incorporate facial recognition technology and enhance their security measures. Deep learning algorithms are being set to ensure that this technology goes beyond regular facial recognition and more understanding images and scenarios. It will also help provide more personalised communications to customers, making it a notable AI trend for coming years.

7. Convergence of IoT and AI

The lines between AI and IoT are increasingly blurring. While both technologies have independent qualities, used together, they are opening up better and more unique opportunities. The Internet of Things (IoT) devices create a lot of data that needs to be mined for actionable insights. On the other hand, Artificial Intelligence algorithms require the data before making any conclusions. So the data collected by IoT is being used by AI algorithms to create valuable results that are further implemented by the IoT devices. AI's ability to rapidly glean insights from data makes IoT systems more intelligent. In upcoming years more than 80% of enterprise IoT projects will incorporate AI in some form, up from just 10% today.

8. AI in Healthcare

The contributions that AI can import to the healthcare industry are working in ground breaking ways, allowing people worldwide to receive safer and more efficient care and making it easier to detect, prevent, and cure diseases. Also, AI's ability to acquire data in real-time from electronic health records, emergency department admissions, equipment utilisation, staffing levels, etc. – and to interpret and analyse it in meaningful ways enables a wide range of efficiency and care-enhancing capabilities in hospital administration. Drug discoveries are another field where AI is acerbating.

AI is playing an essential role in helping healthcare professionals respond to the coronavirus (COVID-19) outbreak. AI is being used to distinguish COVID patients and essential hot spots. COVID vaccine drug discovery is being repurposed and speeded up using AI techniques. Researchers have developed AI-based thermal cameras and smartphone apps for estimating the temperature of people and assembling data for healthcare organisations. Intelligent robots are being deployed to implement "contactless delivery" for isolated individuals, helping medical staff ensure that the key areas stay disinfected and safe for use.

9. Augmented Intelligence

For those who may still be worried about AI cannibalising their jobs, the rise of AI should be a refreshing trend. It brings together the best capabilities of both humans and technology, giving organisations the ability to improve their workforce's efficiency and performance. By 2023, Gartner predicts that 40% of infrastructure and operations teams in large enterprises will use AI-augmented automation, resulting in higher productivity. The healthcare, retail, and travel industries have already created uses of Augmented Reality. Therefore, following this AI trend, there will be an increase in the number of augmented reality apps.

10. Explainable AI

Despite becoming so ubiquitous, AI has suffered from trust issues. Much of what machine learning accomplishes becomes unknowable at various points of the process and appears as a black box. It's often impossible to explain how the AI came to an inevitable conclusion. Explainable AI is designed to simplify and visualise how ML networks make decisions. There is a more significant push for deploying AI in a transparent and clearly defined manner. While companies will make efforts to understand how AI models and algorithms work? AI/ML software providers will make sophisticated ML solutions more explainable to users.

11. Ethical AI

Rising demand for ethical AI is at the top of the list of emerging technology trends. In the past, organisations that adopted Machine Learning and other Artificial Intelligence technologies were not much preoccupied with their ethical impact. Today, however, values-based consumers and employees expect companies to adopt AI responsibly. Over the next few years, firms will deliberately choose to do business with partners that commit to data ethics and adopt data handling practices that reflect their values and customers' values.

More than any other technological future trend, AI future trends promise many possibilities. There are compelling developments; you can't ignore the existence of intelligence

demonstrated by machines. If you are excited to quick start or expand your business solution in the field of AI, the following technology offerings can be the right fit for you: